

Power market strategies

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Three day-ahead strategies I built and backtested on real EPEX auction data (Jan 2019 to Jun 2026), in Python. This is a one-page summary. I have a full two-page tearsheet for each if you want the detail.

Source: EPEX SDAC day-ahead auctions via Energy-Charts (Fraunhofer ISE, CC BY 4.0). Backtest by me. Not investment advice; past performance doesn't indicate future results.

1 • Battery day-ahead arbitrage structural

A 1 MW / 2 MWh battery that buys the cheapest hours and sells the dearest hours of each day's auction. I solve the optimal dispatch as a linear program (HiGHS), with power, capacity, efficiency and cycle limits. The edge is structural: every extra GW of solar deepens the midday dip while the evening peak holds, so the spread a battery captures widens each year. Day-ahead only is the floor; operators stack intraday and ancillary on top, usually another 30 to 80%.

2025 revenue / MW	€85.9k (DE-LU) · €88.4k (CZ)
2026 YTD, annualised	€89.9k / MW
Peak year (2022 gas crisis)	€122.7k / MW
Spec	1 MW / 2 MWh · 86% round-trip · ≤2 cycles/day · fees included

Main risk: edge decay. The spread gets competed away as the storage fleet grows, so revenue per MW falls over time. The annual revenue series is the thing to watch.

2 • Solar cannibalisation spread structural

The same physics as the battery, but without owning one: sell the midday block (11 to 15h), buy the evening block (18 to 21h), equal size, settled at clearing prices. Volume-neutral, so no directional price exposure, just the intraday shape. Solar has to run midday and bids to clear; the evening ramp is met by gas and inflexible demand. As solar grows the midday block gets cannibalised, so the spread widens, and the Mar to Oct solar season is where almost all of it sits.

2025 (Mar to Oct)	+€104/MWh (DE-LU) · +€107/MWh (CZ)
7-year Mar to Oct average	+€70 / MWh
Mar to Oct win rate	100%, four years running
Spec	equal-MW short 11 to 15h / long 18 to 21h · calendar filter, no look-ahead

Main risk: same decay clock as the battery, plus a winter tail. A low-solar evening with a midday price spike inverts it (worst days around minus €85/MWh). Capturing it at size needs exchange or BRP access.

3 • CZ–DE cross-border spread statistical, shown sceptically

CZ and DE-LU are coupled, so their prices are usually identical and only decouple when the interconnector is congested, mostly afternoons under high renewables. The strategy fades those moves back to parity when the daily spread is more than 1.5σ from its 30-day mean. I show it with an honesty check: a "paper" version that trades at the already-printed price (impossible to fill) against a "tradable" version that enters at the next real auction. The lag and costs roughly halve it, and that gap is the point.

Tradable Sharpe (after costs & lag)	1.43
Bootstrap 90% CI	1.1 to 1.8
Paper Sharpe (untradable ceiling)	2.42
P&L / traded day	+€6.8 / MWh

Honest take: even the 1.43 is an upper bound. Paired zonal positions need a BRP in both zones, and liquidity dries up exactly when the spread moves. I treat this as a research lead, not a production strategy.

All figures on realised auction prices, costs included, quoted per unit of position so they scale. Built in Python (pandas, scipy, HiGHS) on ENTSO-E / Energy-Charts / OTE data.